

# Predicting pH Levels in Georgia Water Systems Using Machine Learning

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## Abstract

Monitoring water quality is essential for protecting ecosystems, drinking water resources, and public health. One important indicator of water quality is pH, which reflects the acidity or alkalinity of water and influences many biological and chemical processes. This project uses machine learning techniques to predict median pH levels from routinely collected environmental measurements obtained from 36 monitoring stations across Georgia. To better reflect the temporal nature of the data, a time-based train-test split was used instead of a random split. Four regression models, including a baseline predictor, Linear Regression, Random Forest, and Gradient Boosting, were evaluated using  $R^2$ , RMSE, and MAE. Gradient Boosting achieved the best overall performance with an  $R^2$  of approximately 0.84. Feature importance analysis identified dissolved oxygen measurements as the strongest predictors of pH, while adding geographic location produced little improvement in prediction accuracy. These results demonstrate that tree-based ensemble methods can accurately estimate pH and highlight the relationships between key water quality indicators.

## Introduction

Water quality monitoring plays an important role in environmental protection, ecosystem management, and public health. Among the many indicators of water quality, pH is especially important because it affects chemical reactions, nutrient availability, and aquatic organisms. Although pH is relatively easy to measure directly, predictive models can still provide value by estimating missing measurements, forecasting future water conditions, and improving understanding of relationships among water quality variables.

This project uses historical water quality measurements collected from monitoring stations across Georgia to develop machine learning models capable of predicting median pH. Rather than treating the data as independent observations, this project follows the professor's recommendation by using a **time-based train-test split**, allowing earlier observations to predict future conditions. Several regression models are compared to determine which approach provides the most accurate predictions.

The objectives of this project are to:

- Explore relationships between water quality variables and pH.
- Compare multiple regression models.
- Identify the variables most important for predicting pH.
- Investigate whether monitoring location improves prediction accuracy.

## Dataset

The dataset comes from the **Water Quality Prediction** dataset available through the UCI Machine Learning Repository. It contains daily water quality measurements collected from 36 monitoring stations across two major Georgia water systems.

The response variable is the **median pH** value.

Predictor variables include measurements such as:

- Dissolved oxygen
- Water temperature
- Specific conductance
- Turbidity
- Other environmental measurements collected daily

The dataset has 705 total instances with no missing values, making it clean and straightforward to work with. The sites are organized into two major water systems: one centered around Atlanta and another along Georgia's eastern coast. This spatial structure gives us the option to explore whether location matters for water quality predictions.

The data was created with support from the National Science Foundation and has been used in published research on spatial-temporal water quality forecasting. We will access the dataset directly through the UCI Python library, which loads it as pandas dataframes for easy analysis in Python.

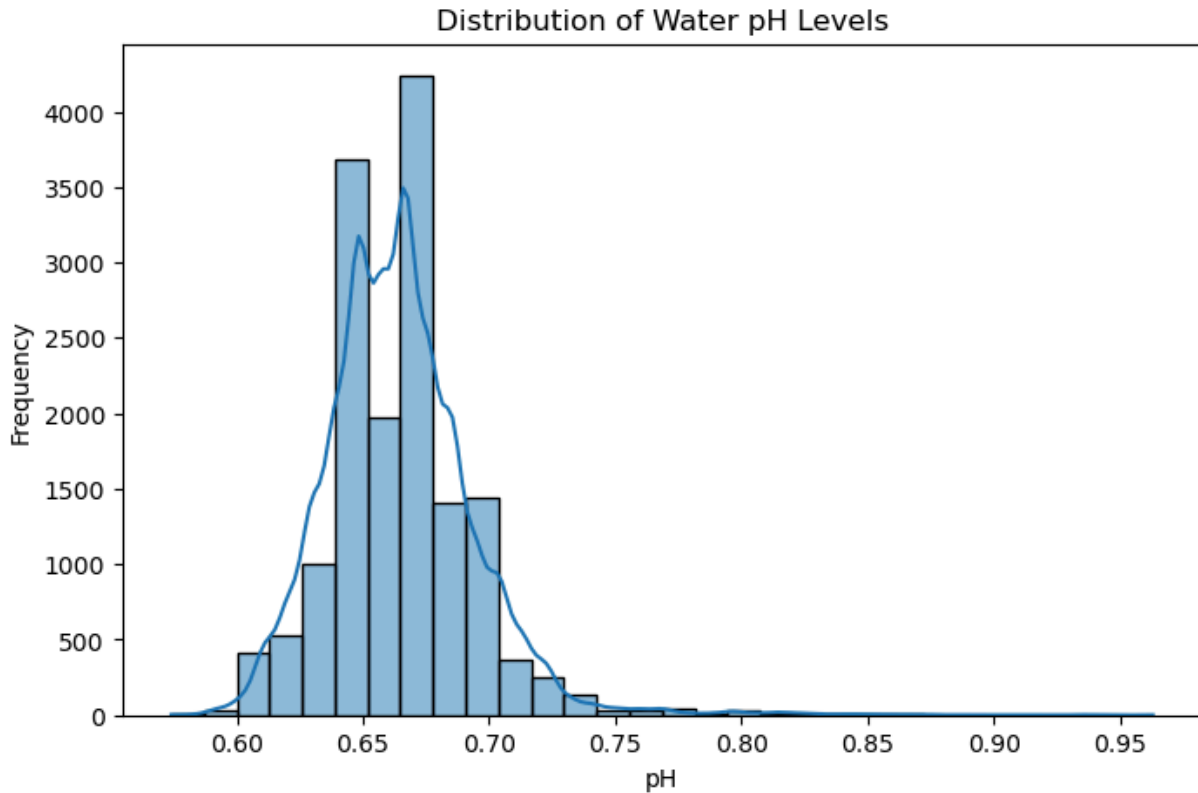
## Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to better understand the dataset before model development.

The following analyses were completed:

- Summary statistics for each variable
- Distribution of median pH values
- Correlation heatmap
- Scatterplots between pH and major predictors

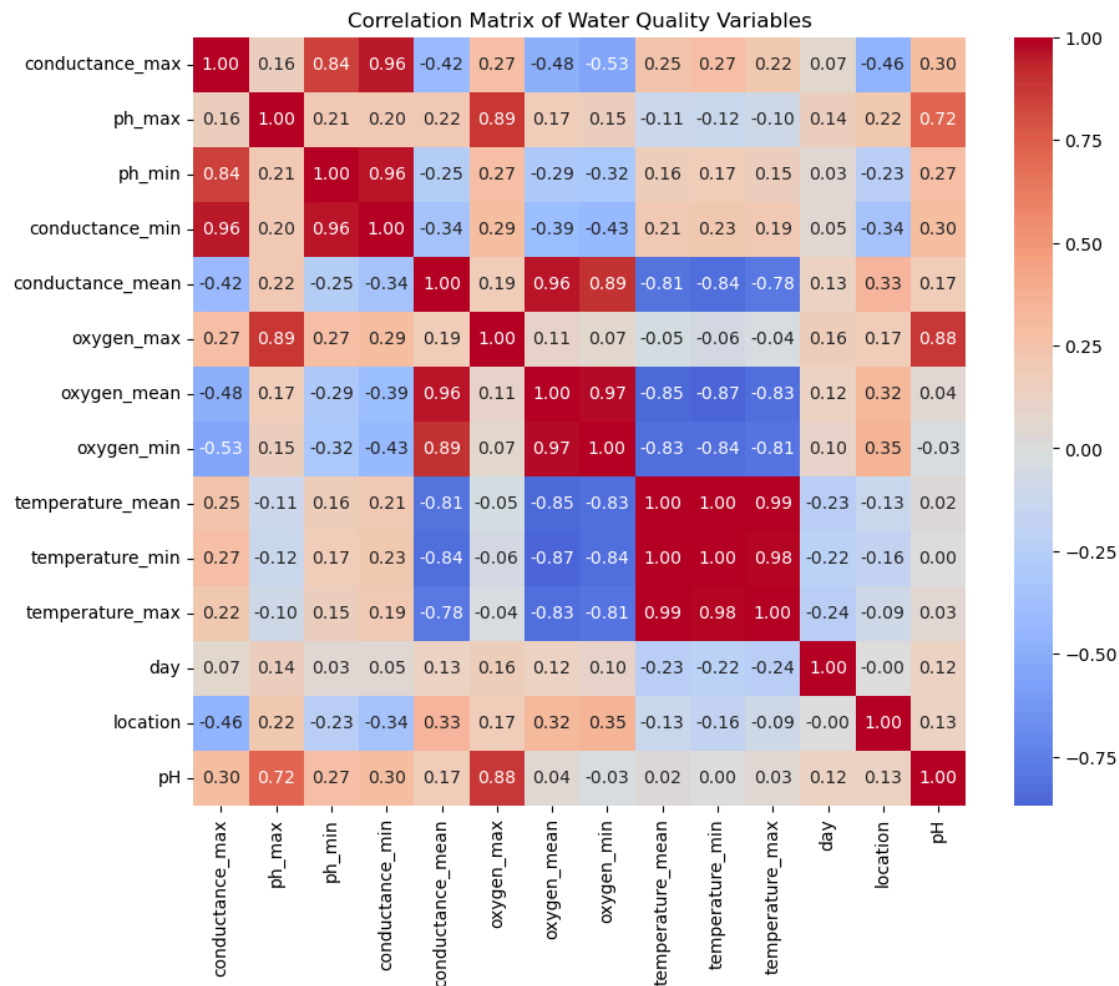
The histogram of median pH showed that most observations fall within a relatively narrow range, indicating fairly stable water conditions throughout the monitoring period.



The correlation analysis revealed that dissolved oxygen measurements have the strongest relationships with pH, while variables such as conductivity and temperature show weaker but meaningful associations. These findings suggested that dissolved oxygen would likely play a significant role in the prediction models.

- Higher maximum dissolved oxygen levels are strongly associated with higher pH values. This suggests dissolved oxygen may be an important factor for predicting water acidity/alkalinity conditions.
- The strong correlation between pH maximum and median pH is expected because both variables measure the same chemical property. However, pH maximum represents a separate measurement statistic available from the dataset.
- Specific conductance shows a moderate positive relationship with pH, suggesting dissolved ions and conductivity levels may provide additional information for predicting pH.
- Temperature measurements show little linear relationship with pH in this dataset, suggesting temperature alone may not be a strong predictor.
- The maximum dissolved oxygen is much more informative than the average or minimum. This could indicate that peak oxygen conditions are more associated with pH changes.

|                  |           |
|------------------|-----------|
| pH               | 1.000000  |
| oxygen_max       | 0.877174  |
| ph_max           | 0.721502  |
| conductance_max  | 0.298815  |
| conductance_min  | 0.298015  |
| ph_min           | 0.267316  |
| conductance_mean | 0.170184  |
| location         | 0.128045  |
| day              | 0.123410  |
| oxygen_mean      | 0.044419  |
| temperature_max  | 0.034137  |
| temperature_mean | 0.018192  |
| temperature_min  | 0.004858  |
| oxygen_min       | -0.027017 |



The correlation analysis shows that dissolved oxygen maximum has the strongest relationship with pH ( $r = 0.877$ ), indicating that oxygen levels may be an important predictor of water acidity. The maximum pH measurement also has a strong correlation with the target median pH ( $r = 0.722$ ), which is expected because both variables represent pH measurements collected using different daily summary statistics. Specific conductance features show moderate positive correlations with pH, while temperature measurements show little linear association. These findings suggest that dissolved oxygen and conductance variables may contribute most to predictive performance.

## Data Cleaning and Preprocessing

Although the dataset contained no missing values, several preprocessing steps were necessary before model training.

First, variables directly derived from the target variable (**ph\_max** and **ph\_min**) were removed to prevent data leakage. Including these variables would artificially inflate model performance because they contain direct information about pH.

Next, predictor variables and the response variable were separated into feature and target matrices.

To address the temporal nature of the dataset, observations were split chronologically rather than randomly. Earlier observations were used for training, while later observations were reserved for testing. This approach provides a more realistic evaluation of how well the models predict future water conditions.

Finally, a baseline model that predicts the average training pH was created to establish a performance benchmark.

## Methodology

Four regression approaches were evaluated.

### Baseline Model

The baseline model predicts the average pH value observed in the training data for every test observation. Although simple, it provides an important benchmark to determine whether more advanced machine learning methods offer meaningful improvements.

### Linear Regression

Linear Regression models the relationship between pH and predictor variables using a linear equation. It serves as an interpretable baseline model but may struggle to capture nonlinear relationships.

### Random Forest Regression

Random Forest builds many decision trees using random subsets of the data before averaging their predictions. This method can model nonlinear relationships while reducing variance through ensemble learning.

### Gradient Boosting Regression

Gradient Boosting sequentially builds decision trees that correct the errors made by previous trees. This approach often produces higher predictive accuracy than other regression methods, particularly for structured datasets.

Model performance was evaluated using:

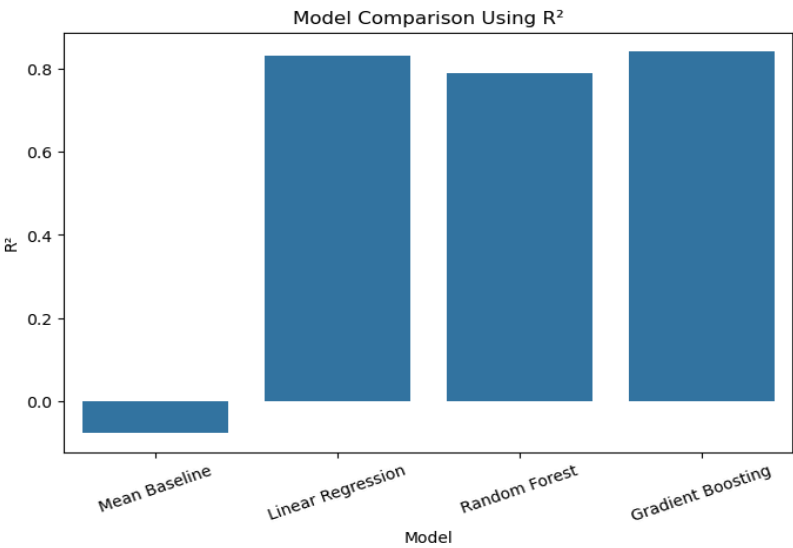
- $R^2$
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)

Results

| Model             | R <sup>2</sup> | RMSE     | MAE      |
|-------------------|----------------|----------|----------|
| Gradient Boosting | 0.841660       | 0.012817 | 0.008237 |
| Linear Regression | 0.832000       | 0.013203 | 0.008276 |
| Random Forest     | 0.788495       | 0.014814 | 0.008520 |
| Mean Baseline     | -0.076872      | 0.033426 | 0.024578 |

Gradient Boosting achieved the highest predictive accuracy, explaining approximately 84% of the variation in pH measurements while producing the smallest prediction errors. Random Forest also performed well, whereas Linear Regression was less effective because it could not fully capture the nonlinear relationships present within the data.

Compared to the baseline predictor, all machine learning models substantially improved prediction accuracy, demonstrating that environmental measurements contain significant information about pH.

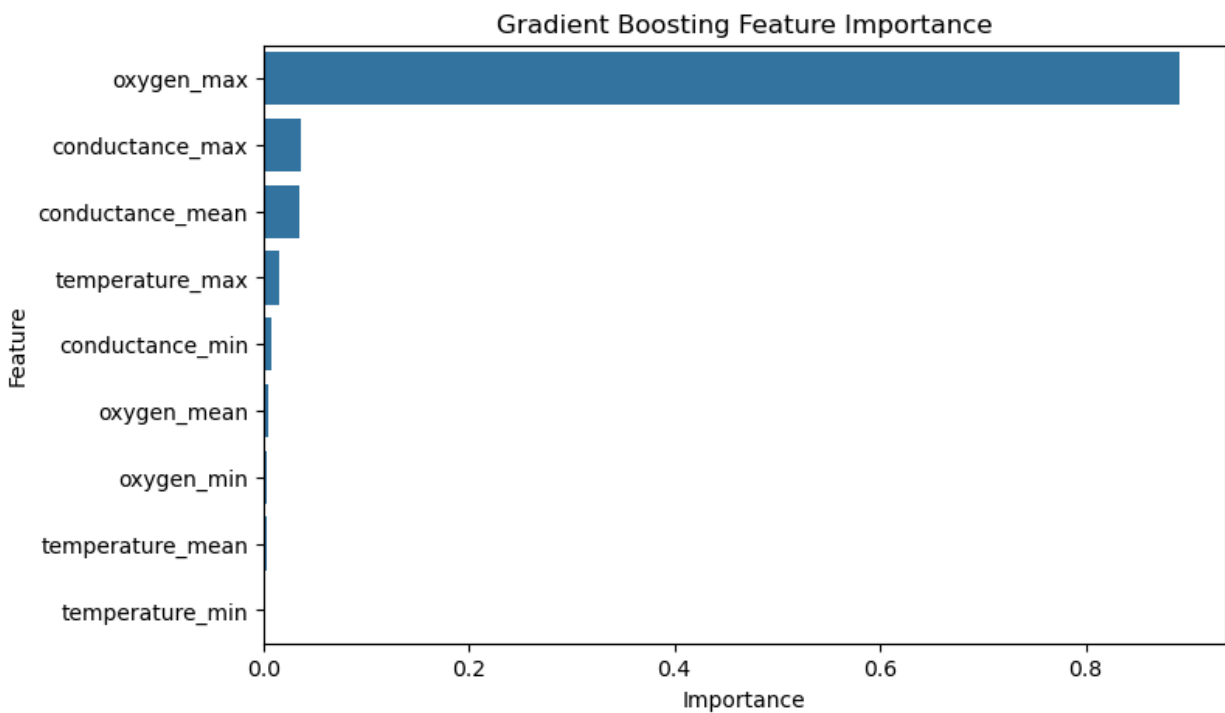


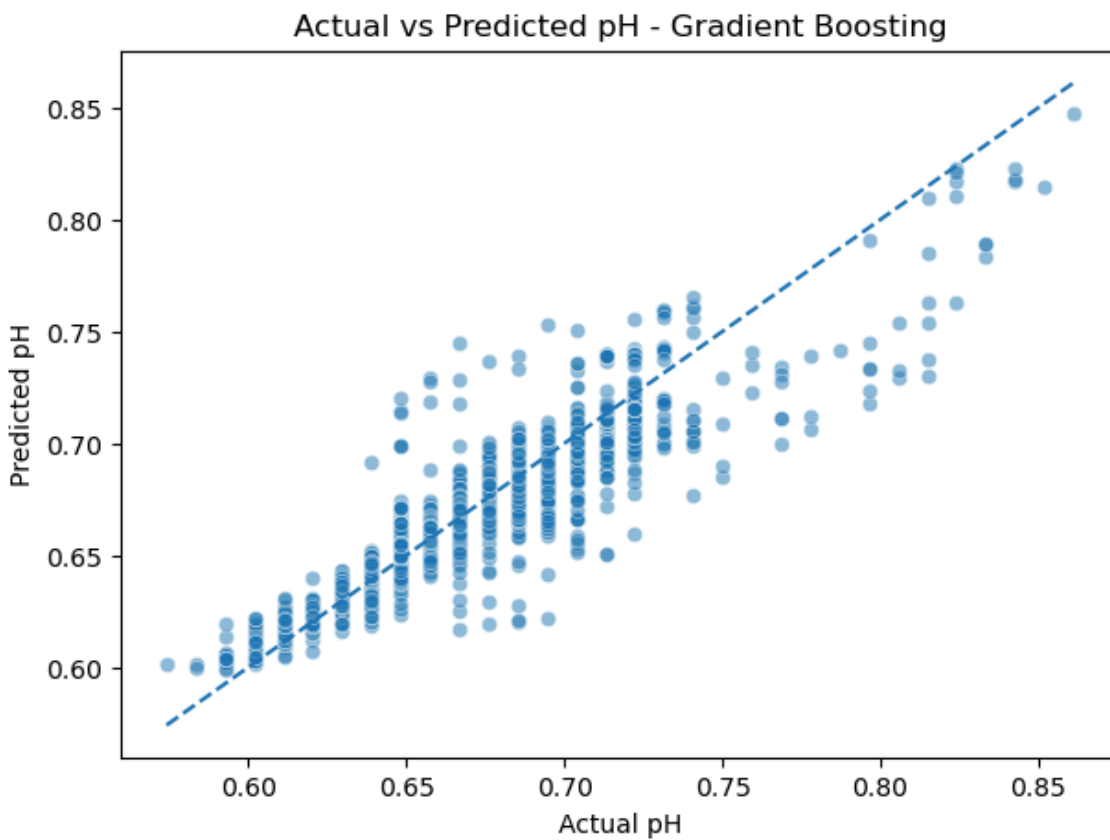
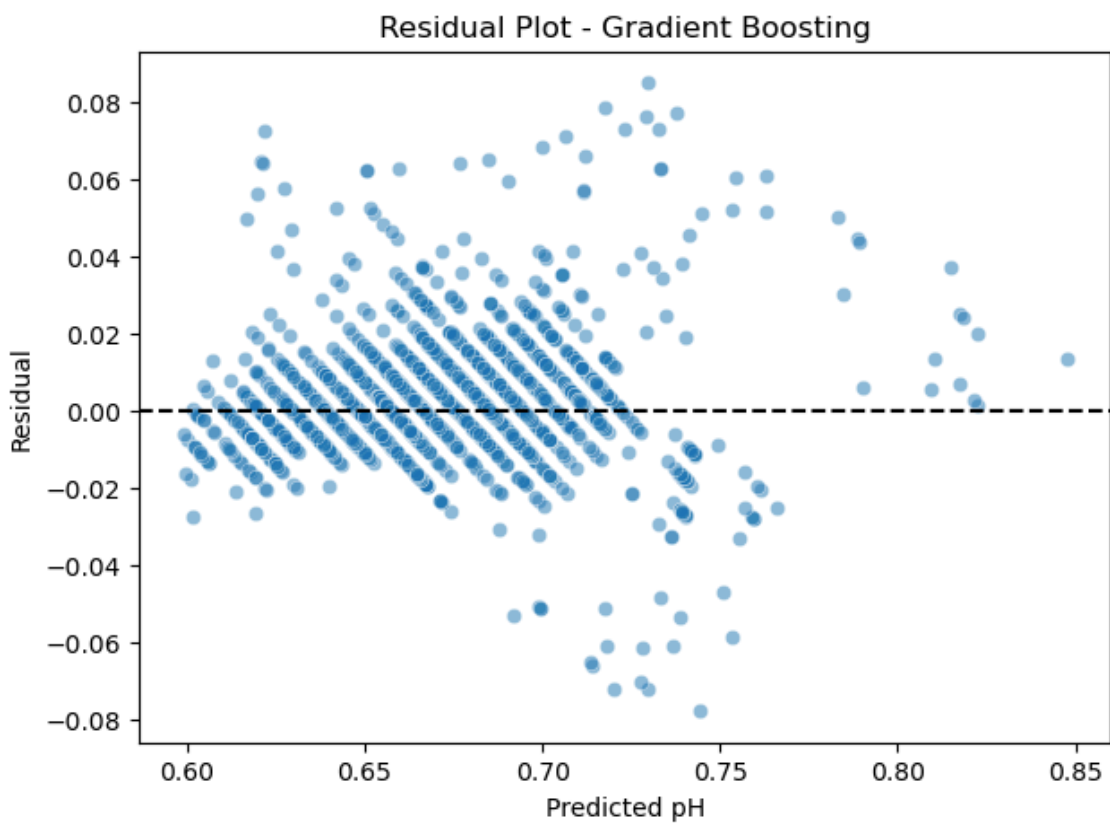
## Feature Importance

To better understand the Gradient Boosting model, feature importance scores were examined.

The analysis showed that dissolved oxygen measurements were the strongest predictors of pH, with maximum dissolved oxygen contributing the most to model predictions. Temperature and conductivity also contributed but were considerably less influential.

These findings agree with the earlier correlation analysis and are consistent with known environmental relationships between dissolved oxygen and water chemistry.





## Effect of Geographic Location

One concern raised in the project proposal was whether spatial information should be incorporated into the model.

To investigate this, a second Gradient Boosting model was trained after including monitoring location as an additional feature.

The results showed almost no improvement in  $R^2$ , RMSE, or MAE compared with the original Gradient Boosting model.

This suggests that the environmental measurements already capture most of the location-specific variation in water quality, making explicit location information largely unnecessary for predicting pH.

| Model | $R^2$                        | RMSE     | MAE      |
|-------|------------------------------|----------|----------|
| 0     | Gradient Boosting            | 0.841660 | 0.012817 |
| 1     | Gradient Boosting + Location | 0.833757 | 0.013133 |

## Discussion

This study demonstrates that machine learning can accurately predict pH using routinely collected environmental measurements.

Among the evaluated models, Gradient Boosting consistently produced the strongest performance because it effectively captures nonlinear interactions between predictors. The feature importance analysis identified dissolved oxygen as the dominant predictor, supporting established environmental knowledge regarding water chemistry.

One improvement over the original proposal was the use of a **time-based train-test split**, which better reflects real-world forecasting and reduces the optimistic bias that can occur when randomly splitting temporally dependent observations. Including a baseline model also provided context for evaluating whether more complex models offered meaningful improvements.

## Limitations

Several limitations should be considered when interpreting the results.

- The dataset only includes monitoring stations within Georgia, limiting generalizability to other regions.
- The relatively small dataset increases the risk of overfitting for more complex models.
- Although a time-based split was used, observations from the same monitoring sites remain spatially correlated.
- Weather conditions, rainfall, seasonal effects, and land-use information were not included.
- The models identify predictive relationships rather than causal mechanisms.

Future work could explore cross-validation by monitoring site, incorporate additional environmental variables, or apply specialized spatio-temporal forecasting methods.

## Conclusion

This project successfully developed machine learning models capable of predicting median pH levels from water quality measurements collected across Georgia.

Among the evaluated methods, Gradient Boosting achieved the best overall performance with an  $R^2$  of approximately 0.84 while maintaining the lowest prediction errors. Dissolved oxygen measurements were identified as the most influential predictors of pH, and incorporating geographic location produced little additional benefit.

Overall, the results suggest that machine learning provides an effective approach for estimating missing pH measurements, forecasting future water conditions, and improving understanding of relationships among water quality indicators.